

DD_RAT-01 - Usage Rating - Developer Implementation Guide

Version: 1.0 Status: Draft for review Bundle: 05_billing Audience: mid-level backend developers, billing developers, QA
Parent DD: DD_RAT-01 - Usage Rating

1. How To Read This Guide

This guide follows the runtime flow.

MED-01 sends a clean mediated usage record. RAT-01 rates it, stores the result, and settles it depending on billing mode.

Do not read raw CDR files in RAT-01. Do not hardcode rates. Do not directly update wallet balances or invoice tables.

2. Step 1 - Receive Mediated Usage

RAT-01 receives UsageMediated:

```
{
  "eventType": "UsageMediated",
  "operatorCode": "WIK",
  "mediatedUsageId": "med-20260602-000001",
  "usageType": "VOICE_CALL",
  "subscriptionId": "SUB-700045",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "serviceRef": "svc_VOIP_STD",
  "packageRef": "pkg_INET_VOICE_STD",
  "calledNumberNormalized": "+254711222333",
  "callDirection": "OUTBOUND",
  "callStartedAt": "2026-06-02T10:15:00+03:00",
  "durationSeconds": 75
}
```

First check idempotency:

```
select *
from rated_usage_record
where mediated_usage_id = 'med-20260602-000001';
```

If a row exists, return or ack without rating again.

3. Step 2 - Resolve Billing Mode

RAT-01 needs to know whether the subscription is PREPAID or POSTPAID at usage time.

Resolution can come from subscription read API or a cached subscription snapshot:

```
{
  "subscriptionId": "SUB-700045",
  "billingMode": "PREPAID",
  "walletId": "WAL-200045",
  "cycleRef": "2026-06"
}
```

If billing mode cannot be resolved:

```
reason_code = BILLING_MODE_NOT_FOUND
```

Create rating quarantine.

4. Step 3 - Lookup Tariff

Call PLM-CFG-07:

```
POST /api/plm/voice-rating/lookup
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "subscriptionId": "SUB-700045",
  "packageRef": "pkg_INET_VOICE_STD",
  "serviceRef": "svc_VOIP_STD",
  "calledNumberNormalized": "+254711222333",
  "callDirection": "OUTBOUND",
  "callStartedAt": "2026-06-02T10:15:00+03:00"
}
```

Response:

```
{
  "tariffPlanCode": "WIK_RES_VOICE_STD_2026",
  "zoneCode": "KE_MOBILE",
  "timeBandCode": "ANYTIME",
  "chargePolicy": "CHARGEABLE",
  "unitType": "MINUTE",
  "unitPrice": 3.0,
  "currency": "KES",
  "initialIncrementSeconds": 60,
  "subsequentIncrementSeconds": 60,
  "setupFeeAmount": 0.0,
  "minimumChargeAmount": 0.0,
  "taxableKind": "SERVICE",
  "taxableRef": "svc_VOIP_STD"
}
```

If lookup returns VOICE_RATE_NOT_FOUND or VOICE_DESTINATION_ZONE_NOT_FOUND, quarantine in RAT-01. MED-01 did its job; the issue is rating configuration.

5. Step 4 - Apply Charge Policy

Charge policy	Action
CHARGEABLE	Continue to allowance, rounding, tax, settlement.
ZERO_RATED	Store rated usage with zero amounts and settlementStatus = NOT_REQUIRED.
BLOCKED	Quarantine with USAGE_BLOCKED_BY_TARIFF.
QUARANTINE	Quarantine with tariff-provided reason.

Zero-rated example:

```
insert into rated_usage_record (
  rated_usage_id,
  mediated_usage_id,
  operator_code,
  usage_type,
  billing_mode,
  subscription_id,
  customer_id,
  service_ref,
  package_ref,
  tariff_plan_code,
  zone_code,
  time_band_code,
  actual_duration_seconds,
  allowance_seconds_applied,
  billable_duration_seconds,
  unit_price,
  gross_amount,
  tax_amount,
  net_amount,
  currency_code,
  rating_status,
  settlement_status,
  rated_at
) values (
  'rated-20260602-000002',
  'med-20260602-000002',
  'WIK',
  'VOICE_CALL',
  'PREPAID',
  'SUB-700045',
  'CUS-100045',
  'svc_VOIP_STD',
  'pkg_INET_VOICE_STD',
```

```

'WIK_RES_VOICE_STD_2026',
'EMERGENCY',
'ANYTIME',
30,
0,
0,
0,
0,
0,
0,
0,
'KES',
'ZERO_RATED',
'NOT_REQUIRED',
now()
);

```

6. Step 5 - Apply Allowance

Load eligible allowances from PLM-CFG-07 response or allowance config cache.

If an allowance applies, lock bucket:

```

select *
from usage_allowance_bucket
where operator_code = 'WIK'
  and subscription_id = 'SUB-700045'
  and allowance_code = 'ONNET_100_MIN'
  and cycle_ref = '2026-06'
for update;

```

If bucket does not exist, create it from PLM-CFG-07 allowance definition.

Example:

```

insert into usage_allowance_bucket (
  bucket_id,
  operator_code,
  subscription_id,
  tariff_plan_code,
  allowance_code,
  cycle_ref,
  included_seconds,
  consumed_seconds,
  remaining_seconds,
  status,
  created_at,
  updated_at
) values (
  'uab-202606-sub700045-onnet',
  'WIK',
  'SUB-700045',
  'WIK_RES_VOICE_STD_2026',
  'ONNET_100_MIN',
  '2026-06',
  6000,
  0,
  6000,
  'ACTIVE',
  now(),
  now()
);

```

Consumption example:

```

actual duration = 75 seconds
allowance remaining = 60 seconds
allowance applied = 60 seconds
paid seconds before rounding = 15 seconds

```

Update bucket:

```

update usage_allowance_bucket
set consumed_seconds = consumed_seconds + 60,
  remaining_seconds = remaining_seconds - 60,
  status = case when remaining_seconds - 60 <= 0 then 'EXHAUSTED' else status end,
  updated_at = now()
where bucket_id = 'uab-202606-sub700045-onnet';

```

Allowance consumption must happen in the same transaction as rated usage creation or with a compensating rollback if settlement fails before final commit.

7. Step 6 - Round Billable Duration

Use tariff increment fields.

For 75 seconds with 60/60 increment and no allowance:

```
first 60 seconds -> 60 billed
remaining 15 seconds -> round to next 60
billable duration = 120 seconds
```

Pseudo-code:

```
function billableSeconds(int $paidActualSeconds, int $initialIncrement, int $subsequentIncrement): int
{
    if ($paidActualSeconds <= 0) {
        return 0;
    }

    if ($paidActualSeconds <= $initialIncrement) {
        return $initialIncrement;
    }

    $remaining = $paidActualSeconds - $initialIncrement;
    $roundedRemaining = (int) ceil($remaining / $subsequentIncrement) * $subsequentIncrement;

    return $initialIncrement + $roundedRemaining;
}
```

8. Step 7 - Calculate Gross Amount

For unit type MINUTE:

```
billable_seconds = 120
unit_price = 3.00 KES per minute
billed_minutes = 120 / 60 = 2
gross_amount = 2 * 3.00 = 6.00 KES
```

Then apply setup fee and minimum charge:

```
gross_amount = max(gross_amount + setup_fee_amount, minimum_charge_amount)
```

9. Step 8 - Compute Tax

If gross_amount > 0, call PLM-CFG-02:

```
POST /api/tax/compute
Content-Type: application/json

{
    "operatorCode": "WIK",
    "taxableKind": "SERVICE",
    "taxableRef": "svc_VOIP_STD",
    "baseAmount": 6.00,
    "currency": "KES",
    "customerId": "CUS-100045",
    "subscriptionId": "SUB-700045",
    "taxableAt": "2026-06-02T10:15:00+03:00"
}
```

Example response:

```
{
    "taxAmount": 0.96,
    "totalAmount": 6.96,
    "taxLines": [
        {
            "taxType": "VAT",
            "rate": 0.16,
            "baseAmount": 6.00,
            "taxAmount": 0.96
        }
    ]
}
```

Store the full tax breakdown in `rated_usage_record.tax_breakdown_json`.

10. Step 9 - Store Rated Usage

Insert immutable rated usage:

```
insert into rated_usage_record (
  rated_usage_id,
  mediated_usage_id,
  operator_code,
  usage_type,
  billing_mode,
  subscription_id,
  customer_id,
  account_id,
  service_ref,
  package_ref,
  tariff_plan_code,
  zone_code,
  time_band_code,
  actual_duration_seconds,
  allowance_seconds_applied,
  billable_duration_seconds,
  unit_price,
  gross_amount,
  tax_amount,
  net_amount,
  currency_code,
  tax_breakdown_json,
  rating_status,
  settlement_status,
  rated_at
) values (
  'rated-20260602-000001',
  'med-20260602-000001',
  'WIK',
  'VOICE_CALL',
  'PREPAID',
  'SUB-700045',
  'CUS-100045',
  'ACC-900045',
  'svc_VOIP_STD',
  'pkg_INET_VOICE_STD',
  'WIK_RES_VOICE_STD_2026',
  'KE_MOBILE',
  'ANYTIME',
  75,
  0,
  120,
  3.000000,
  6.000000,
  0.960000,
  6.960000,
  'KES',
  '["taxType": "VAT", "rate": 0.16, "baseAmount": 6.00, "taxAmount": 0.96]',
  'RATED',
  'PENDING',
  now()
);
```

Emit UsageRated.

11. Step 10 - PREPAID Wallet Settlement

If billing mode is PREPAID and net_amount > 0, call BIL-05:

```
POST /api/wallets/WAL-200045/debit
Authorization: Bearer <internal-service-token>
Idempotency-Key: RAT:med-20260602-000001
Content-Type: application/json

{
  "amount": 6.96,
  "type": "DEBIT_USAGE_VOICE",
  "idempotencyKey": "RAT:med-20260602-000001",
  "reference": "rated-20260602-000001",
  "actor": "service:rat-01",
  "metadata": {
    "mediatedUsageId": "med-20260602-000001",
```

```

        "usageType": "VOICE_CALL",
        "calledNumberMasked": "+254711***333",
        "billableDurationSeconds": 120
    }
}

```

Success response:

```

{
  "walletTransactionId": "wtx-20260602-000001",
  "walletId": "WAL-200045",
  "newBalance": 993.04,
  "status": "POSTED"
}

```

Update rated usage:

```

update rated_usage_record
set settlement_status = 'SETTLED',
    settlement_ref_id = 'wtx-20260602-000001'
where rated_usage_id = 'rated-20260602-000001';

```

Record settlement attempt in usage_settlement_attempt.

12. Step 11 - Handle PREPAID Insufficient Balance

If BIL-05 returns:

```

{
  "errorCode": "INSUFFICIENT_BALANCE",
  "currentBalance": 3.00,
  "shortfall": 3.96
}

```

Do not delete the rated usage. Update:

```

update rated_usage_record
set settlement_status = 'FAILED'
where rated_usage_id = 'rated-20260602-000001';

```

Record attempt:

```

target_module = BIL-05
status = FAILED_FINAL
error_code = INSUFFICIENT_BALANCE

```

Emit PrepaidUsageDebitFailed.

Downstream dunning/restriction policy should consume this event or read wallet state. RAT-01 does not decide service restriction.

13. Step 12 - POSTPAID Rated Usage

If billing mode is POSTPAID, do not debit wallet.

Store:

```

settlement_status = POSTPAID_PENDING_INVOICE

```

Emit PostpaidUsageReadyForInvoice.

Invoice generation reads:

```

GET /api/usage-rating/postpaid-usage?subscriptionId=SUB-700045&cycleRef=2026-06

```

Example response:

```

{
  "items": [
    {
      "ratedUsageId": "rated-20260602-000010",
      "serviceRef": "svc_VOIP_STD",
      "zoneCode": "KE_MOBILE",
      "billableDurationSeconds": 120,
      "grossAmount": 6.00,
      "taxAmount": 0.96,
      "netAmount": 6.96,
      "currency": "KES"
    }
  ]
}

```

```

    ],
    "totals": {
      "grossAmount": 6.00,
      "taxAmount": 0.96,
      "netAmount": 6.96,
      "currency": "KES"
    }
  }
}

```

After invoice generation includes these records:

```

POST /api/usage-rating/postpaid-usage/mark-invoiced
Content-Type: application/json

{
  "invoiceId": "INV-2026-06-000045",
  "ratedUsageIds": ["rated-20260602-000010"]
}

```

RAT-01 updates those rows to INVOICED.

14. Step 13 - Rating Quarantine

If tariff lookup fails:

```

insert into usage_rating_quarantine (
  rating_quarantine_id,
  mediated_usage_id,
  operator_code,
  reason_code,
  severity,
  status,
  context_json,
  created_at
) values (
  'urq-20260602-000001',
  'med-20260602-000099',
  'WIK',
  'VOICE_RATE_NOT_FOUND',
  'HIGH',
  'OPEN',
  '{"calledNumber": "+998123456", "packageRef": "pkg_INET_VOICE_STD"}',
  now()
);

```

After catalog config is fixed, Backoffice calls:

```

POST /api/usage-rating/quarantine/urq-20260602-000001/replay

```

RAT-01 re-runs rating using the same mediated_usage_id.

15. Suggested Worker Commands

These are Laravel commands or queue jobs and can be consolidated into billing worker pods.

```

rat:consume-mediated-usage
rat:rate-pending
rat:retry-settlement
rat:replay-quarantine --id=urq-20260602-000001
rat:expire-allowance-buckets
rat:postpaid-cycle-readiness-check --cycle=2026-06

```

Recommended schedule:

Command	Frequency
rat:consume-mediated-usage	Continuous Kafka consumer.
rat:rate-pending	Every minute if polling MED-01.
rat:retry-settlement	Every 5 minutes for retryable target failures.
rat:replay-quarantine	Manual or event-driven after catalog repair.
rat:expire-allowance-buckets	Daily.
rat:postpaid-cycle-readiness-check	Before invoice generation.

16. Error Codes

Error code	Meaning
RAT_MEDIATED_USAGE_NOT_FOUND	MED-01 record cannot be loaded.
RAT_DUPLICATE_MEDIATED_USAGE	Record already rated.
RAT_BILLING_MODE_NOT_FOUND	Subscription billing mode cannot be resolved.
RAT_TARIFF_LOOKUP_FAILED	PLM-CFG-07 lookup failed.
RAT_RATE_NOT_FOUND	No applicable rate exists.
RAT_TAX_COMPUTE_FAILED	PLM-CFG-02 tax compute failed.
RAT_WALLET_NOT_FOUND	PREPAID subscription has no wallet.
RAT_WALLET_DEBIT_FAILED	BIL-05 debit failed.
RAT_INSUFFICIENT_BALANCE	BIL-05 returned insufficient balance.
RAT_POSTPAID_ALREADY_INVOICED	Attempt to mark usage invoiced twice.

17. Minimum Implementation Order

Build in this order:

1. Migrations for rated usage, allowance bucket, allowance consumption, rating quarantine, settlement attempt, rating job.
2. Consumer for UsageMediated.
3. Idempotency check on mediated_usage_id.
4. Subscription billing-mode resolver.
5. PLM-CFG-07 rating lookup client.
6. Duration rounding calculator.
7. Allowance bucket create/consume logic.
8. PLM-CFG-02 tax compute client.
9. Rated usage writer.
10. PREPAID BIL-05 debit client.
11. POSTPAID usage read and mark-invoiced APIs.
12. Rating quarantine and replay APIs.
13. Settlement retry worker.
14. Backoffice read APIs.

18. Test Scenarios

18.1 Prepaid Voice Call

Input:

- Mediated record duration 75 seconds.
- Rate 3 KES/minute, 60/60.
- Tax 16%.
- Wallet has enough balance.

Expected:

- Rated gross 6.00.
- Tax 0.96.
- Net 6.96.

- BIL-05 debit called once.
- Rated usage SETTLED.

18.2 Duplicate Mediated Event

Input:

- Same mediatedUsageId received twice.

Expected:

- One rated usage record.
- One wallet debit.

18.3 Allowance Applies

Input:

- 75-second call.
- Remaining allowance 60 seconds.

Expected:

- Allowance consumption 60 seconds.
- Paid actual seconds 15.
- Billable paid seconds 60 with 60/60 increment.

18.4 Zero-Rated Emergency Call

Input:

- Tariff lookup returns ZERO_RATED.

Expected:

- Rated usage stored with zero amounts.
- No wallet debit.

18.5 Missing Tariff

Input:

- PLM-CFG-07 returns no rate.

Expected:

- Rating quarantine RAT_RATE_NOT_FOUND.
- No wallet debit.

18.6 Insufficient Wallet Balance

Input:

- Wallet balance below net amount.

Expected:

- Rated usage exists.
- Settlement status FAILED.
- PrepaidUsageDebitFailed emitted.

18.7 Postpaid Usage

Input:

- POSTPAID subscription.

Expected:

- Rated usage POSTPAID_PENDING_INVOICE.
- Invoice generation can read it.
- Mark-invoiced changes status to INVOICED.

19. Developer Checklist

Before marking RAT-01 complete:

- No raw CDR parsing exists in RAT-01.
- Rates are looked up from PLM-CFG-07.
- Tax is computed by PLM-CFG-02.
- mediated_usage_id uniqueness prevents double rating.
- Allowance bucket updates are locked and atomic.
- Rated usage is immutable after creation.
- Wallet debit uses stable idempotency key.
- Settlement attempts are recorded.
- Postpaid invoice handoff prevents double invoicing.
- Rating quarantine and replay work.